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B.Tech. Degree VII Semester Examination November 2017

IT 1705 (E2) CRYPTOGRAPHY AND DATA SECURITY (2012 Scheme)

Time : 3 Hours

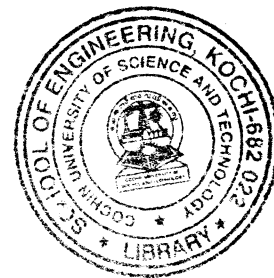
Maximum Marks : 100

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) Define cryptology? Explain the two areas of cryptology.
- (b) Explain the different types of cryptanalytic attacks.
- (c) Differentiate between stream and block enciphering.
- (d) Summarize the characteristics of DES algorithm.
- (e) Differentiate between asymmetrical and symmetrical cipher systems with examples.
- (f) Describe Hash and DES hash with figures.
- (g) Explain the general aspects of key management.
- (h) Describe end to end encipherment with figure.



PART B

(4 × 15 = 60)

- II. (a) Explain the different aspects of information measure and absolute security. (10)
- (b) Write short note on Kasiski test. (5)
- OR**
- III. (a) Explain the working of Hagelin machine with an example. (8)
- (b) Describe the various classical cipher systems. (7)
- IV. Describe DES algorithm with steps and diagrams. (15)
- OR**
- V. (a) Describe the various modes of DES. (9)
- (b) Explain IDEA. (6)
- VI. What do you mean by public key system? Explain any public key system with example. (15)
- OR**
- VII. Describe message authentication with MAC and digital signature. (15)
- VIII. (a) Explain different types of encipherment implemented in a network for key distribution. (10)
- (b) Describe Diffie-Hellman key exchange. (5)
- OR**
- IX. Differentiate how key is distributed for asymmetrical and symmetrical systems. (15)